INCH-POUND
MIL-PRF-8805/45H
25 May 2022
SUPERSEDING
MIL-PRF-8805/45G
31 January 2006

PERFORMANCE SPECIFICATION SHEET

# SWITCHES, SENSITIVE, SPST, 5 AMPERES, RESILIENT SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8805.

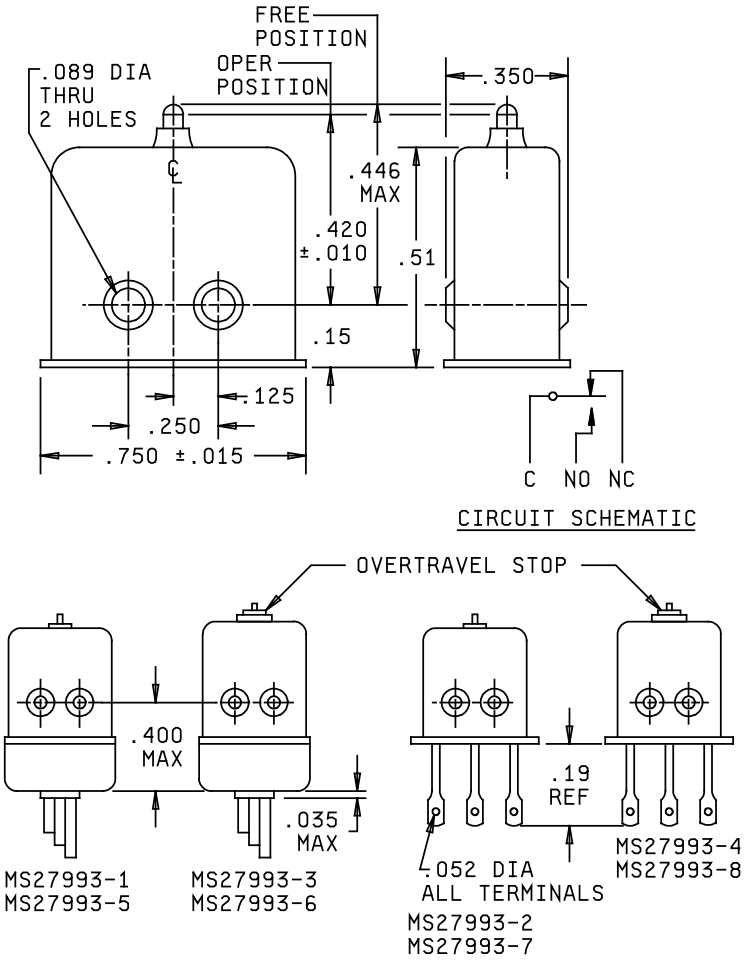



FIGURE 1. Dimensions and configuration.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .005 | 0.13 | .089 | 2.26 | .350 | 8.89 |
| .010 | 0.25 | .125 | 3.18 | .400 | 10.16 |
| .015 | 0.38 | .130 | 3.30 | .420 | 10.67 |
| .020 | 0.51 | .150 | 3.81 | .446 | 11.33 |
| .035 | 0.89 | .190 | 4.83 | .510 | 12.95 |
| .052 | 1.32 | .250 | 6.35 | .750 | 19.05 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (0.51 mm) for two place decimals and ±.005 (.13 mm) for 3 place decimals.
4. Switch configuration optional provided dimensions specified are not exceeded.
5. Insulated conductors in accordance with M5086/1-22 per SAE-AS5086/1 or equivalent, 18 inches long.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 4 (resilient).

Temperature characteristic:

| | |
|---|---|
| MS27993-1: | 1 (- 55°C to + 85°C). |
| MS27993-2: | (- 55°C to + 125°C). |
| MS27993-3: | 1 (- 55°C to + 85°C). |
| MS27993-4: | (- 55°C to + 125°C). |
| MS27993-5: | 1 (- 55°C to + 85°C). |
| MS27993-6: | 1 (- 55°C to + 85°C). |
| MS27993-7: | (- 55°C to + 125°C). |
| MS27993-8: | (- 55°C to + 125°C). |

Shock type: M (100 g, test condition I, MIL-STD-202-213).

Vibration grade: 2 (10 Hz to 2,000 Hz ).

Weight: See table I.

Operating characteristics: See table I.

Contact resistance: 80 milliohms maximum.

Dielectric withstanding voltage:

Sea level: 1,250 V rms for one second.
Seal test: Only water tight test shall be performed in group A. Switches need not be removed from underwater during test.

Mechanical endurance:

MS27993-1 thru –4 100,000 cycles at a rate of 10 to 12 cycles per minute.
MS27993-5 thru –8 100,000 cycles.

Electrical endurance: 50,000 cycles.

Electrical ratings: See table II.

Part or Identifying Number (PIN): See table I.

Quality assurance: Group B inspection – See table IV.



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TABLE I. PIN's and operating characteristics.

| PIN | Actuating force | Releasing force | Movement differential | Over travel | Weight | Weight with wire | Applied actuator force | Termination |
|---|---|---|---|---|---|---|---|---|
| MS27993-1 | 22 oz. Max. | 6 oz. Min. | .0005 to .005 inch | .007 inch max. 1/ | ----- | .0616 pound max. | Does not apply | Lead wire |
| MS27993-2 | 22 oz. Max. | 6 oz. Min. | .0005 to .005 inch | .007 inch max. 1/ | .02 pound max. | ----- | Does not apply | Solder |
| MS27993-3 | 22 oz. Max. | 6 oz. Min. | .0005 to .005 inch | .006 inch min. | ----- | .0616 pound max. | 10 pounds max | Lead wire |
| MS27993-4 | 22 oz. Max. | 6 oz. Min. | .0005 to .005 inch | .006 inch min. | .02 pound max. | ----- | 10 pounds max | Solder |
| MS27993-5 | 10 oz. Max. | 3 oz. Min. | .0005 to .005 inch | .007 inch max. 1/ | ----- | .0616 pound max. | Does not apply | Lead wire |
| MS27993-6 | 10 oz. Max. | 3 oz. Min. | .0005 to .005 inch | .006 inch min. | ----- | .0616 pound max. | 10 pounds max | Lead wire |
| MS27993-7 | 10 oz. Max. | 3 oz. Min. | .0005 to .005 inch | .007 inch max. 1/ | .02 pound max. | ----- | Does not apply | Solder |
| MS27993-8 | 10 oz. Max. | 3 oz. Min. | .0005 to .005 inch | .006 inch min. | .02 pound max. | ----- | 10 pounds max | Solder |

1/ Overtravel stop not provided. In use, do not exceed.

TABLE II. Electrical ratings.

| Load | Sea level | | 50,000 feet |
|---|---|---|---|
| | 28 V dc | 115 V ac 400 Hz | 28 V dc |
| | Amperes | Amperes | Amperes |
| Resistive | 5 | 5 | 5 |
| Inductive | 3 | 3 | 3 |
| Lamp | 1.5 | 1.5 | 1.5 |



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TABLE III. Qualification group submission.

| Examination or test | Basic switch | Other switch samples | Extent of approval |
|---|---|---|---|
| Qualification inspection table of MIL-PRF-8805 Groups I. through VIII. Inclusive | MS27993-8<br>(All sample units) | MS27993-1<br>MS27993-2<br>MS27993-3<br>MS27993-4<br>MS27993-5<br>MS27993-6<br>MS27993-7<br><br>(3 sample units)<br><br>Visual and mechanical inspection | All |

TABLE IV. Group B inspection.

| Examination or test | Basic switch | Other switches |
|---|---|---|
| Group B inspection table of MIL-PRF-8805 group I. thru IV. | MS27993-8<br>(All sample units) | MS27993-1<br>MS27993-2<br>MS27993-3<br>MS27993-4<br>MS27993-5<br>MS27993-6<br>MS27993-7<br><br>Visual and mechanical inspection |



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Referenced Documents
MIL-PRF-8805
MIL-STD-202-213
SAE-AS5086/1

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2022-049)

Review activities:
Army - AR, AV, CR4, MI
Navy – AS, MC, OS, SH

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